

Asheesh Kumar

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Education

ABES Engineering College	2023 – 2027
B.Tech in Information Technology	CGPA: 7.94
Jawahar Navodaya Vidyalaya	2021 – 2022
Class XII	88%
Jawahar Navodaya Vidyalaya	2019 – 2020
Class X	88.6%

Technical Skills

- **Generative AI & LLMs:** LangChain (LCEL), LangGraph (State Machines, Checkpointing, Human-in-the-Loop), RAG Architecture, Vector DBs (pgvector, FAISS), Agentic Tool-Calling, Sentence-Transformers, Pydantic Structured Output, Prompt Engineering, LangSmith (Tracing/Observability)
- **Machine Learning & NLP:** Scikit-learn, XGBoost, Feature Engineering, EDA, Text Preprocessing, Tokenization, TF-IDF, Attention Mechanism, Transformers
- **Deep Learning Frameworks:** TensorFlow, Keras, PyTorch (Basics), ANN, CNN, RNN, LSTM, Sequence Modeling
- **Data & Databases:** PostgreSQL (pgvector), SQLite, SQL, Pandas, NumPy, Relational Database Modeling
- **Deployment & Tools:** FastAPI, Streamlit, Docker, Docker Compose, REST APIs, Git, GitHub, Linux, VS Code
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks

Projects

Enterprise Hybrid-RAG Engine GitHub — Live Demo

Python, LangChain (LCEL), PostgreSQL (pgvector), FastAPI, Docker, Streamlit

- Architected a containerized Hybrid-RAG engine using **PostgreSQL (pgvector)** with **HNSW indexing**, enabling sub-millisecond semantic retrieval across 384-dimensional dense embeddings (Sentence-Transformers).
- Implemented a declarative **LangChain LCEL** pipeline with **Pydantic structured output parsing**, generating grounded answers with verified confidence scores (0.0–1.0), verbatim citation quotes, and interactive follow-ups.
- Built an autonomous **tool-calling agent** for dynamic routing between enterprise vector search, live database metadata statistics, and DuckDuckGo web search fallback.
- Implemented **Hybrid Search** (dense cosine similarity + sparse full-text search) and orchestrated the multi-service system (Postgres, FastAPI, Streamlit) using **Docker Compose**, optimizing the production image from 10GB to 1.2GB via CPU-only dependency pruning.

Quora Duplicate Question Detection GitHub — Live Demo

Python, Scikit-learn, XGBoost, Sentence-Transformers, FastAPI, Streamlit

- Built an end-to-end NLP pipeline to detect duplicate question pairs on the Quora Question Pairs dataset (404K labeled pairs; trained on a 30K sample), engineering **20+ features**: token/length overlap, **TF-IDF**, and fuzzy-matching scores implemented from scratch with **difflib**.
- Compared **Logistic Regression**, **Random Forest** and **XGBoost** on accuracy, F1 and **log-loss** (the Kaggle metric); an ablation showed that adding a **Sentence-Transformers** cosine-similarity feature lifted XGBoost from **78.2% to 84.1% accuracy** and cut log-loss from 0.423 to 0.342 on a held-out split.
- Served the best model through a **FastAPI** endpoint (Pydantic-validated) and a **Streamlit** demo sharing a single inference module; added unit tests for the preprocessing and feature code.

Emotion Classification NLP (BiGRU + FastAPI) GitHub — Live Demo

Python, TensorFlow/Keras, FastAPI, Pydantic, HTML/CSS/JS

- Designed and trained a **BiGRU** deep learning model (Embedding → BiGRU → Dense → Softmax) to classify text into **six emotion classes** (joy, sadness, anger, fear, love, surprise).
- Achieved **92.25% test accuracy**, **88.61% macro F1-score**, and **92.45% weighted F1-score** through tokenization, sequence padding, and embedding-based preprocessing.
- Deployed the model as a **FastAPI** REST service (with auto-generated Swagger docs) returning the predicted emotion, confidence score, and full class-probability distribution; built a lightweight HTML/CSS/JS frontend for interactive live predictions.

Achievements

- Solved **330+** Data Structures and Algorithms problems on LeetCode with a maximum rating of **1600**.
- Hands-on expertise across Data Structures, Generative AI (RAG, Agents, LangGraph), Machine Learning, and Full-Stack API deployment.